UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

STAT 576 - Empirical Process Theory and Weak Convergence Fall 2025
Problem Set 4 due November 7, 2025

Exercise 1. Let $(\mathcal{F}_k)_{k=1}^n$ be a filtration and suppose Δ_k is a martingale increment with respect to this filtration, i.e. for every k, Δ_k is $\mathcal{F}_k$-measurable and $\mathbb{E}[\Delta_k \mid \mathcal{F}_{k-1}] = 0$. Furthermore, suppose for every k we have random variables A_k and B_k which are $\mathcal{F}_{k-1}$-measurable such that $A_k \le \Delta_k \le B_k$ almost surely.

Prove the following inequality:

$$\mathbb{P}\left[\sum_{k=1}^n \Delta_k \ge t \text{ and } \sum_{k=1}^n (B_k - A_k)^2 \le c^2\right] \le e^{-2t^2/c^2}.$$

Note that this generalizes Azuma–Hoeffding because we do not assume $|B_k - A_k|$ is uniformly bounded. As a hint, consider the quantity

$$\lambda \sum_{k=1}^n \Delta_k - \frac{\lambda^2}{8} \sum_{k=1}^n (B_k - A_k)^2.$$

Exercise 2. In the lecture, we have established the concentration of the MaxCut value for the $\mathbb{G}(n, dn)$ random graph model. The models $\mathbb{G}(n, dn)$ and $\mathbb{G}(n, c/n)$ are in some appropriate sense equivalent when $d = c/2$. Towards seeing this fact,

(a) Establish that the random number of edges in $\mathbb{G}(n, c/n)$ concentrates around its expectation w.h.p. as $n \to \infty$. State and prove this fact rigorously.

(b) Establish the concentration of the MaxCut value for the $\mathbb{G}(n, c/n)$ random graph model.

Exercise 3. An *m-clique* of a graph is a subgraph on m vertices with all $\binom{m}{2}$ edges present—that is, a complete graph on m vertices. Let N_m denote the number of m-cliques in the random graph $\mathbb{G}(n, p)$ for a constant $p \in (0, 1)$.

(a) For every constant m, establish a bound of the form

$$\mathbb{P}\big(|N_m - \mathbb{E}[N_m]| \ge \varepsilon \mathbb{E}[N_m]\big) \le \exp(-\Omega(n^\alpha))$$

for the largest possible exponent α you can obtain.

(b) Show that the rate α obtained in part (a) is tight.

Exercise 4. Given n points $x_1, \ldots, x_n \in [0,1]^2$, consider the set of all permutations on these points. Each such permutation results in a circuit or loop when the first point in the permutation is appended at the end (resulting in a sequence of length $n+1$). Denote the set of all such circuits by $\mathcal{S}_n$ (clearly, $|\mathcal{S}_n| = n!$). For $\pi(n+1) := \pi(1)$, define:

$$T(x_1, \ldots, x_n) \;=\; \min_{\pi \in \mathcal{S}_n} \sum_{i=1}^{n} \|x_{\pi(i)} - x_{\pi(i+1)}\|.$$

Let $X_1, \ldots, X_n$ be i.i.d. random variables distributed uniformly in $[0,1]^2$, and let $Z = T(X_1, \ldots, X_n)$. Show that Z is tightly concentrated around its mean.

Exercise 5. Let V be a normed space, and $v_1, v_2, \ldots, v_n \in V$ such that $|v_i| \leq 1$ for all $i = 1, \ldots, n$. Set $X = |\sum_{i=1}^{n} \epsilon_i v_i|$, where $\epsilon_1, \ldots, \epsilon_n$ are i.i.d. Rademacher variables. Prove that for any $0 < \delta < 1$, we have with probability at least $1 - \delta$:

$$X \leq \mathbb{E}[X] + \sqrt{2n \ln \frac{1}{\delta}}.$$

Hint: Define a Doob martingale and apply Azuma's inequality.

Exercise 6. Let $X_1, \ldots, X_n$ be n i.i.d copies of X that has cumulative distribution function $F(t) = \mathbb{P}(X \leq t)$. The empirical CDF of X is defined by

$$\hat{F}_n(t) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}(X_i \leq t).$$

1. Compute the mean and the variance of $\hat{F}_n(t)$ and conclude that $\hat{F}_n(t) \to F(t)$ as $n \to \infty$ almost surely (hint: use Borel-Cantelli).

2. Show that for $n \geq 2$

$$\sup_{t \in \mathbb{R}} |\hat{F}_n(t) - F(t)| \leq C \sqrt{\frac{\log(n/\delta)}{n}}$$

with probability $1 - \delta$.

Exercise 7 (Population minimizers). In empirical risk minimization for prediction based on pairs (X, Y), we target the minimum of the population risk over some function class $\mathcal{G}$. Here we focus on the largest choice, which is the set of all measurable functions $x \mapsto g(x) \in \mathbb{R}$ and study the properties of the minimizer $g^* := \inf_g \bar{R}(g)$ where $\bar{R}(g) := \mathbb{E}_{X,Y}[\mathcal{L}(g(X), Y)]$.

Compute the function g^* for the following choices. (Assume that all necessary expectations exist.)

(a) Regression on scalar outputs with the weighted least-squares loss

$$\mathcal{L}(g(x), y) = w(x)(y - g(x))^2$$

for some positive weight function $x \mapsto w(x) > 0$.

(b) Regression with $y \in \mathbb{R}$ and the tilted absolute value

$$\mathcal{L}_\alpha(g(x), y) = \begin{cases} \alpha|y - g(x)|, & \text{if } y \geq g(x), \\ |y - g(x)|, & \text{otherwise} \end{cases}$$

for some scalar $\alpha > 0$.

(c) Binary classification with $y \in \{0, 1\}$ and the logistic loss

$$\mathcal{L}(g(x), y) = -g(x)y + \log(1 + e^{g(x)}).$$

(d) Binary classification with $y \in \{0, 1\}$ and the binary Kullback-Leibler (KL) loss for function values $g(x) \in [0, 1]$:

$$\mathcal{L}(g(x), y) = y \log\left(\frac{y}{g(x)}\right) + (1 - y) \log\left(\frac{1 - y}{1 - g(x)}\right).$$

Hint: Note that we are trying to recover $g^*(x)$ for all x, which is equivalent to optimizing the conditional expectation $\mathbb{E}_{X,Y}\big[\mathcal{L}(g(X), Y) \mid X = x\big]$.

Exercise 8. Consider the function class

$$\mathcal{F} = \{x \mapsto \operatorname{sign}(\langle \theta, x \rangle) \mid \theta \in \mathbb{R}^d, \|\theta\|_2 = 1\},$$

corresponding to the $\{-1, +1\}$-valued classification rules defined by linear functions in $\mathbb{R}^d$. Supposing that $d \geq n$, let $x_1^n = \{x_1, \ldots, x_n\}$ be a collection of vectors in $\mathbb{R}^d$ that are linearly independent. Show that the empirical Rademacher complexity satisfies

$$\mathcal{R}(\mathcal{F}(x_1^n)/n) = \mathbb{E}_\varepsilon \left[\sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(x_i) \right| \right] = 1.$$

Discuss the consequences for empirical risk minimization over the class $\mathcal{F}$.

Exercise 9. Prove the following properties of the Rademacher complexity.

(a) $\mathcal{R}_n(\mathcal{F}) = \mathcal{R}_n(\operatorname{conv}(\mathcal{F}))$, where conv denotes the *convex hull*.

(b) Show that $\mathcal{R}_n(\mathcal{F} + \mathcal{G}) \leq \mathcal{R}_n(\mathcal{F}) + \mathcal{R}_n(\mathcal{G})$. Give an example to demonstrate that this bound cannot be improved in general.

(c) Given a fixed and uniformly bounded function g, show that

$$\mathcal{R}_n(\mathcal{F} + g) \leq \mathcal{R}_n(\mathcal{F}) + \|g\|_\infty / \sqrt{n}.$$